AVISHKAAR

Team Name

“Do not include any organization or college logos. Only your startup logo or team logo may be added — no other logos or symbols are allowed.”

Problem Statement

Solution

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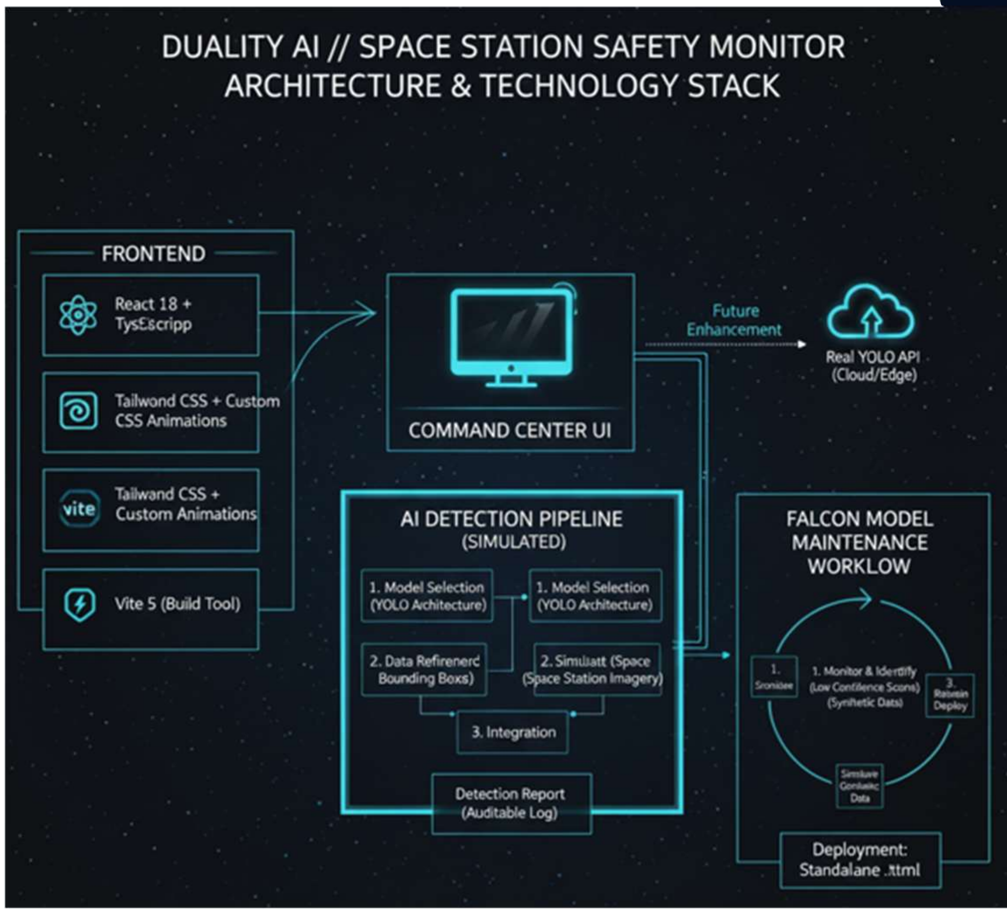
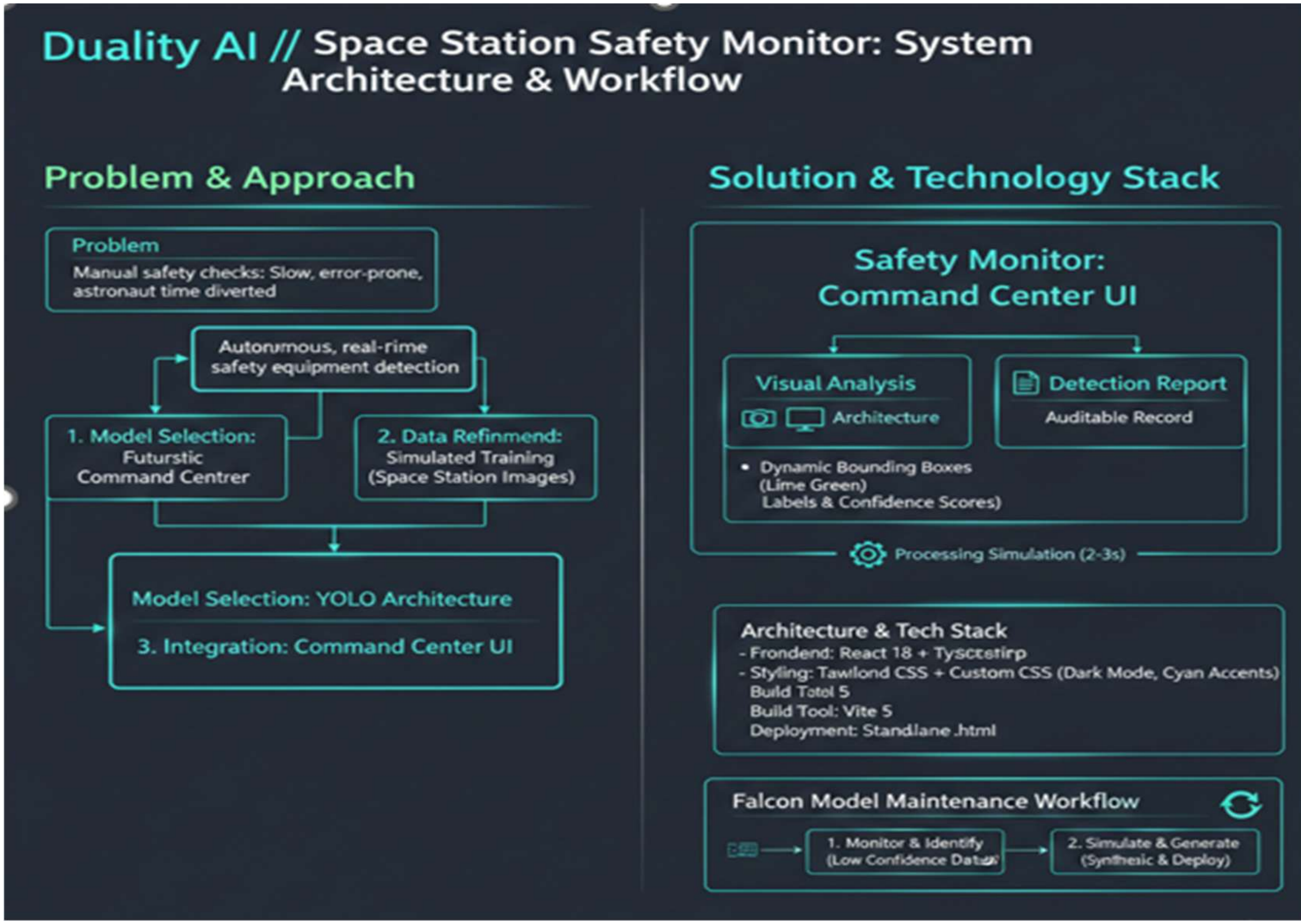
The Duality AI // Space Station Safety Monitor employs a clean, single-page application architecture designed for performance and responsiveness. The frontend is robustly built using React 18 with TypeScript, providing a modern, scalable, and type-safe foundation for the entire Command Center UI. For styling and user experience, we've leveraged Tailwind CSS combined with custom CSS animations, creating a distinct dark mode interface accented with glowing cyan elements and a starfield background, truly embodying a futuristic aesthetic. Development efficiency and optimized build sizes are ensured by Vite 5, our chosen build tool. A key advantage of this stack is its deployment simplicity: the application can be easily deployed as a standalone .html file, requiring no complex server dependencies, making it highly portable and accessible for critical space station environments.



At the core of the system lies a simulated AI Detection Pipeline, which conceptualizes the integration of a YOLO-based object detection model. This pipeline demonstrates the process from model selection (adopting YOLO's renowned architecture for speed and accuracy) and simulated data refinement (training on diverse space station imagery), through to integration within our interactive Command Center UI. This simulation effectively validates the core concept, visualization, and user experience for monitoring the seven critical safety equipment classes. Furthermore, to address long-term model viability, we've outlined the Falcon Model Maintenance Workflow, a strategy for continuous monitoring, synthetic data generation, and retraining/redeployment, ensuring the system's adaptability and sustained accuracy against real-world changes and model drift.



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The Duality AI // Space Station Safety Monitor tackles the critical issue of manual safety checks in space. Our AI-powered system, simulated with a YOLO model, autonomously detects essential equipment like fire extinguishers and oxygen tanks within a futuristic command center UI. This responsive interface, built with React, TypeScript, and Tailwind CSS, provides both visual analysis with dynamic bounding boxes and an auditable detection report. The Falcon Workflow ensures continuous model accuracy. This project highlights a robust, autonomous solution for enhanced space station safety.

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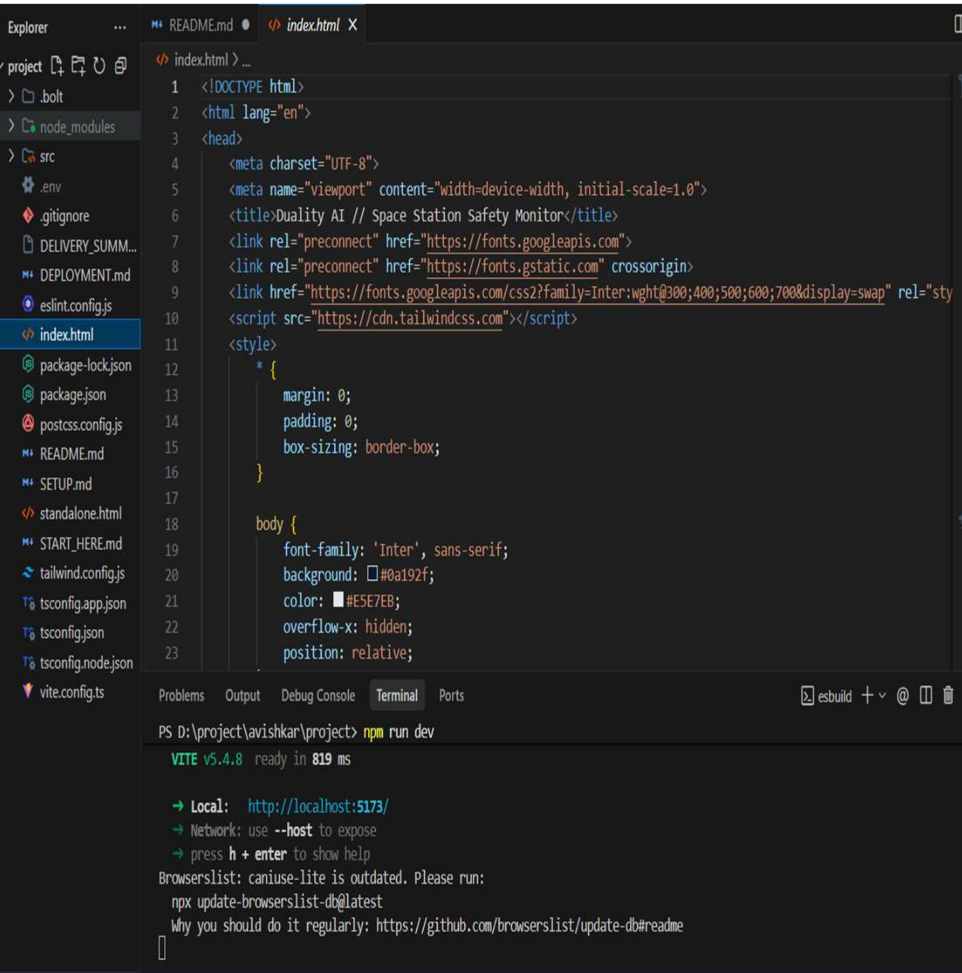
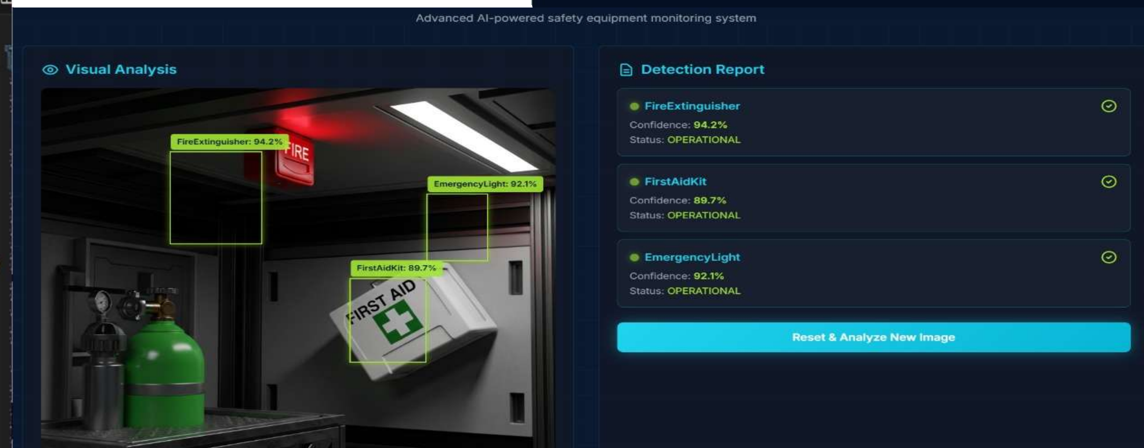


Explorer
README.md
index.html
project
.bolt
node_modules
src
.env
.gitignore
DELIVERY_SUMM...
DEPLOYMENT.md
eslint.config.js
index.html
package-lock.json
package.json
postcss.config.js
README.md
SETUP.md
standalone.html
START_HERE.md
tailwind.config.js
tsconfig.app.json
tsconfig.json
tsconfig.node.json
vite.config.ts
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Duality AI // Space Station Safety Monitor</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap" rel="sty
<script src="https://cdn.tailwindcss.com"></script>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}
body {
font-family: 'Inter', sans-serif;
background: #0a192f;
color: #E5E7EB;
overflow-x: hidden;
position: relative;
Problems
Output
Debug Console
Terminal
Ports
esbuild
PS D:\project\avishkar\project> npm run dev
VITE v5.4.8 ready in 819 ms
➜ Local: http://localhost:5173/
➜ Network: use --host to expose
➜ press h + enter to show help
Browserslist: caniuse-lite is outdated. Please run:
npx update-browserslist-db@latest
Why you should do it regularly: https://github.com/browserslist/update-db#readme



Advanced AI-powered safety equipment monitoring system
Visual Analysis
FireExtinguisher: 94.2%
EmergencyLight: 92.1%
FirstAidKit: 89.7%
FIRST AID
Detection Report
FireExtinguisher
Confidence: 94.2%
Status: OPERATIONAL
FirstAidKit
Confidence: 89.7%
Status: OPERATIONAL
EmergencyLight
Confidence: 92.1%
Status: OPERATIONAL
Reset & Analyze New Image

Duality AI // Space Station Safety Monitor
Safety Detection Interface
Advanced AI-powered safety equipment monitoring system
Drag & Drop Test Image
or click to upload
Supported formats: JPG, PNG
Model Maintenance via Falcon